Structure, Not Vocabulary: A Contrastive-Pairs Diagnostic for Political Manipulation Detection

Working Paper III in Computational Political Anthropology

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Code & data: github.com/politomorphism/politomorphism-defense

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Abstract

Detectors trained on propaganda corpora can flag charged vocabulary without representing the rhetorical structure that makes a text manipulative. Working Papers I and II compared a lexical baseline and a fine-tuned transformer on an article-disjoint re-split of the English PropXplain corpus, but neither could separate structural sensitivity from a lexical correlate. This paper introduces a contrastive-pairs diagnostic to make that separation. Each item is a pair of short political sentences on the same topic, matched in vocabulary (mean member Jaccard 0.50), where one member carries exactly one manipulative device — presupposition, manufactured equivalence, false dilemma, whataboutism, a loaded rhetorical question, or implied causation — and the other states the same content plainly. Because the words are held nearly constant, a model that scores the manipulative member higher must respond to structure. On sixty such pairs (ten per technique), the TF-IDF + logistic-regression baseline reaches a pair accuracy of 0.633 [95% CI 0.51, 0.76], barely above the chance value of 0.50, while the fine-tuned DistilBERT reaches 0.917 [0.85, 0.99]; the confidence intervals do not overlap, and a paired test on the identical pairs confirms the difference (exact McNemar $p = 0.0002$). The gap is concentrated where the lexical model fails structurally — false dilemma (0.30 to 1.00) and loaded rhetorical questions (0.00 to 0.70). Across five training seeds the transformer margin is stable (mean pair accuracy 0.880, 95% CI [0.852, 0.908]; every seed above the baseline). An independent model rater (OpenAI gpt-4o-mini), blind to the construction, labels the 120 members with substantial agreement to the author labels (Cohen's kappa 0.617; pairwise 0.917), and its errors fall on different techniques than the detectors', which argues that the pairs probe distinct structures rather than one vocabulary artefact. A residual-lexical analysis confirms the margin is structural: the transformer's advantage is uncorrelated with the pairs' lexical overlap ($r = +0.04$, n.s.), whereas the lexical baseline's success is significantly correlated with it ($r = +0.50$, $p < 0.001$). The instrument is offered as a diagnostic with fully open materials, not as a validated measure of manipulation; its remaining limits — a model rather than a human second rater, residual vocabulary difference, and author-constructed stimuli — define the next steps.

Keywords: construct validity; contrastive evaluation; propaganda detection; rhetorical structure; manipulation; DistilBERT; inter-rater agreement; computational political anthropology; Politomorphism

1. The question: structure or vocabulary?

A text is politically manipulative not because of the words it uses but because of what it does with them — presupposing a contested claim as settled, forcing a false either/or, deflecting a criticism by pointing at the other side. The same charged vocabulary can appear in a plain, honest sentence, and a

manipulative structure can be built from mild words. Any detector meant to support a defensive audit of political communication must therefore answer a prior question about itself: when it flags a passage as propaganda, is it responding to the rhetorical structure, or merely to a vocabulary that correlates with propaganda in its training corpus? The two are easy to confuse, because in natural corpora charged words and manipulative structures travel together. A detector can score well on a held-out split of the same corpus while having learned only the lexical correlate, and that distinction is invisible to any evaluation carried out on the corpus itself.

This is a construct-validity question in the classical sense: does the instrument measure the construct it claims to measure, or a confound (Cronbach & Meehl, 1955)? The two working papers that precede this one could not answer it. Working Paper I built a transparent lexical baseline on an article-disjoint re-split of the English PropXplain corpus; Working Paper II showed that a fine-tuned transformer classifies the same labels better at equal review cost. Both results are about corpus-label transfer — predicting the corpus's propaganda labels — and both are silent on whether either model represents manipulation as a structure. This paper builds an instrument designed specifically to separate the two, and reports what it finds.

2. Method: contrastive pairs

The method is to hold vocabulary approximately fixed and vary only structure. Each item is a pair of short sentences on the same topic. One member, the manipulative member, introduces exactly one named rhetorical device; the other, the control, states the same content plainly — as an open question, an attributed claim, or a multi-option statement — removing the device without adding another. The two members are written to share as many content words as possible, so that the difference between them is structural, not lexical. A model that assigns a higher propaganda score to the manipulative member of a well-matched pair must be responding to something beyond the word list.

A valid pair satisfies five construction rules: same topic; matched vocabulary; exactly one structural device in the manipulative member; a plain control that removes the device without introducing a new one; and a written one-sentence rationale recording why each member is what it is. Six devices are covered, each drawn from the standard taxonomy of propaganda and manipulation techniques (Da San Martino et al., 2020) and from pragmatics: presupposition (a contested proposition encoded as given), manufactured equivalence (two unlike things asserted as identical), false dilemma (a choice collapsed to two options), whataboutism (a criticism deflected to the opponent), loaded rhetorical question (a question that asserts its own answer), and implied causation (causation inferred from mere sequence or co-occurrence). Grounding the device definitions in published taxonomy, rather than private judgment, is deliberate: it makes each label an application of a public definition that others can contest.

The primary metric is pair accuracy: the fraction of pairs in which the manipulative member receives the higher propaganda score. On perfectly matched pairs a purely lexical model should sit near the chance value of 0.50; a model sensitive to structure should sit well above it. Two controls guard the reading. The first is a lexical-overlap control: for each pair we report the Jaccard overlap of the two members' word sets, so that a high pair accuracy on well-matched pairs cannot be dismissed as the model keying on different words. The second is the comparison between a lexical baseline and a contextual model: the decisive quantity is not either model's absolute pair accuracy but the margin by which the contextual model exceeds the lexical one on matched pairs — the margin that, if positive and robust, is the evidence that context buys structural sensitivity.

3. Materials and models

The benchmark was built in three stages and is released at all three. A 24-pair pilot (v0.1) proved the method but was too loosely matched (mean member Jaccard 0.21). Version 0.2 rewrote it as 30 tightly matched pairs (Jaccard 0.53). Version 0.3, used for all results below, expands it to 60 pairs — ten per technique — at a mean member Jaccard of 0.50, so roughly half the words in each pair are shared between the manipulative and control member. Two models score every member. The lexical baseline is the TF-IDF + logistic-regression model of Working Paper I (lowercased, unigrams and bigrams, document-frequency floor, sublinear term frequency, class-balanced). The contextual model is the DistilBERT of Working Paper II, fine-tuned on the clean article-disjoint PropXplain split. Each model assigns a propaganda score to each of the 120 members; pair accuracy follows directly.

4. Results

4.1 The lexical baseline is near chance; the contextual model is not

On the 60 vocabulary-matched pairs the result is clean and points one way. The lexical baseline reaches a pair accuracy of 0.633 [95% CI 0.51, 0.76] — only a little above the 0.50 a model that cannot tell structure apart would score — while the fine-tuned transformer reaches 0.917 [0.85, 0.99]. The two confidence intervals do not overlap, so the margin of about +0.28 is statistically distinguishable, not merely suggestive. Table 1 breaks this down by technique.

Technique	Pairs	Baseline	Transformer
presupposition	10	1.00	1.00
manufactured equivalence	10	0.90	0.90
false dilemma	10	0.30	1.00
whataboutism	10	0.80	1.00
loaded rhetorical question	10	0.00	0.70
implied causation	10	0.80	0.90
ALL	60	0.633 [.51,.76]	0.917 [.85,.99]

Table 1. Pair accuracy by technique: the manipulative member scored higher. Mean member Jaccard 0.50. Chance is 0.50.

More telling than the overall number is where the gap lives. On the two techniques where the lexical baseline fails structurally once vocabulary is matched — loaded rhetorical questions (baseline 0.00, transformer 0.70) and false dilemma (0.30 to 1.00) — the contextual model recovers most or all of the signal. Because the two members of each pair share most of their words, that recovery cannot be attributed to vocabulary; it is direct evidence that the contextual model is sensitive to rhetorical structure, which is the claim the corpus-level studies could not establish. Where both models already succeed (presupposition, manufactured equivalence), plain-word cues are enough and the two agree.

Because both models were scored on the identical sixty pairs, the comparison can be made as a paired test rather than as two separate intervals. The paired analysis is decisive. The transformer is correct on nineteen pairs where the baseline is wrong, against only two pairs where the baseline is right and the transformer wrong (exact McNemar test, $p = 0.0002$); the two models are both correct on thirty-six pairs and both wrong on three. A bootstrap confidence interval on the paired margin — the transformer's pair accuracy minus the baseline's, resampled over pairs — is [+0.15, +0.42], excluding zero. The advantage is therefore not an artefact of two confidence intervals that happen not to touch; it is a significant, paired difference on the same items.

4.2 The margin is stable across training seeds

A single training run could be lucky. The transformer was therefore retrained five times, holding the data split fixed and varying only the model's initialisation and data order; every run scored the same 60 pairs. Table 2 reports the five pair accuracies.

Seed	Pair accuracy	vs baseline 0.633
13	0.833	above
21	0.900	above
42	0.917	above
101	0.867	above
202	0.883	above
mean	0.880 (sd 0.032)	95% CI [0.852, 0.908]

Table 2. Transformer pair accuracy across five training seeds, fixed split. Every seed exceeds the lexical baseline; the across-seed interval excludes it.

Every individual seed sits far above the lexical baseline, and the across-seed 95% confidence interval [0.852, 0.908] excludes it entirely, so the margin is a stable property of the contextual model, not an accident of initialisation. The per-technique ranking is also stable across seeds: false dilemma, implied causation, presupposition, and manufactured equivalence average 0.92 to 0.96, while loaded rhetorical question remains the weakest at 0.62 (range 0.50 to 0.70) — the one place where the residual difference in control phrasing still costs the model.

4.3 An independent rater recovers the labels

The pairs and their manipulative/control labels were written by the author. To check that they are not one person's private judgment, an independent rater that took no part in building them — a separate large language model, OpenAI gpt-4o-mini — labelled every sentence blind. The 120 members were shuffled so that no pair was ever seen side by side, and each was judged alone as manipulative or plain. Table 3 summarises the agreement.

Independent rater (gpt-4o-mini), blind	Value
Per-member raw agreement with author labels	80.8% (97/120)
Cohen's kappa	0.617 (substantial)
Recall on manipulative (sensitivity)	0.85
Recall on control (specificity)	0.77
Pairwise forced choice (picks author's manipulative)	55/60 = 0.917

Table 3. Agreement of an independent, blind model rater with the author labels. Kappa band after Landis & Koch (1977).

An outside model, blind to the construction, recovers the manipulative/plain distinction well above chance (Cohen's kappa 0.617, in the substantial band of Landis & Koch, 1977), so the labels are not one person's private judgment. This is a model rater, not a human one, and is reported as such: it strengthens the labels, it does not certify them. Its disagreements are structured rather than random, and they reinforce the construct rather than undermine it. The rater's fourteen false alarms — a control called manipulative — fall mostly on manufactured equivalence, where the control keeps charged words (“dictatorship”, “thief”) that lure a model keying on vocabulary; its nine misses — a manipulative called plain — are all presuppositions, the subtlest device, which is hard to see in a

sentence read in isolation. Crucially, these are not the techniques on which the detectors failed: the lexical baseline failed on loaded rhetorical questions and false dilemma, while the independent rater is perfect on both and stumbles elsewhere. That the failures move around by technique, from rater to rater, is itself evidence that the pairs probe distinct rhetorical structures rather than one vocabulary artefact that every model would catch or miss together.

4.4 The transformer's margin does not track vocabulary; the baseline's success does

Because the pairs are matched but not word-identical (mean Jaccard 0.50), a sceptic can still argue that the contextual model wins on the residual vocabulary difference. The scores already in hand test that directly. If the transformer were exploiting leftover words, its advantage would grow as the two members diverge lexically — it would win most on poorly matched pairs and stumble where the words are nearly identical. The opposite holds. The transformer's score margin is essentially uncorrelated with lexical overlap (Pearson $r = +0.04$, $p = 0.75$), and its pair accuracy is, if anything, slightly higher on the well-matched half (0.93, Jaccard above the median) than on the poorly matched half (0.83) — the reverse of what a vocabulary-exploiting model would show. As a control, the lexical baseline behaves exactly as a lexical model should: its correctness is significantly correlated with overlap (point-biserial $r = +0.50$, $p < 0.001$), rising from 0.43 on poorly matched pairs to 0.83 on well-matched ones. On the strict high-overlap subset (Jaccard ≥ 0.60, twenty pairs, where almost all words are shared), both models score high (transformer 0.93, baseline 0.90), as expected when only a few structural words remain to carry the signal. The pattern across the full range is the informative one: the lexical model's success depends on how much vocabulary distinguishes the members, and the contextual model's does not. That is direct evidence that the transformer's margin comes from rhetorical structure rather than from the vocabulary left unmatched.

5. Discussion

Four independent lines converge. On matched vocabulary, a lexical model sits near chance and a contextual model does not; the gap survives retraining across five seeds; an outside rater, blind to the design, recovers the same distinction with substantial agreement; and the contextual model's margin, unlike the lexical model's success, does not track the pairs' residual vocabulary difference. Together these support a narrow but real claim: on this instrument, the contextual model is sensitive to rhetorical structure, not merely to vocabulary, and the manipulative/plain distinction the pairs encode is recoverable by a rater who never saw how they were built. This is precisely the claim that an evaluation carried out on the training corpus cannot make, because there structure and vocabulary are confounded. It is also the claim a defensive audit layer needs before it can be trusted: an instrument that flags manipulation should respond to the manipulation, not to a vocabulary that happens to accompany it.

The claim is deliberately modest in scope. Pair accuracy measures sensitivity to rhetorical structure with vocabulary held approximately fixed; it does not measure whether a passage “is propaganda” in any political or legal sense, whether it would persuade a real audience, or whether the construct “manipulation” is validly operationalised in full. A high pair accuracy is evidence that a model represents a structure; it is not evidence that the model, or the corpus labels it was trained on, captures manipulation as it operates in the world. Construct validity is earned incrementally — technique by technique, rater by rater — and this paper reports one increment, not a verdict.

6. Limitations

Four limits bound the result, and they are stated plainly because the programme's credibility rests on honest scoping. First, the independent check is a model, not a human; a human annotator and an inter-annotator agreement among people remain the stronger evidence and the natural next step,

and the model rater's own substantial-but-imperfect agreement (kappa 0.617) shows the labels are defensible, not settled. Second, mean member overlap is 0.50 — substantial but not identity — so a residue of vocabulary difference remains, concentrated on loaded rhetorical questions whose control phrasing ("people are asking whether …") differs in form; the weakest technique is exactly the one where that residue is largest. Third, the pairs are author-constructed stimuli, not naturally occurring text; they isolate structure by design, which is their purpose, but a model's success on them does not guarantee the same sensitivity on messy real-world passages. Fourth, the contextual model is one modestly sized transformer trained on one corpus; the result shows that context can buy structural sensitivity here, not that transformers "detect manipulation" in general.

7. Ethical note

This instrument scores rhetorical structure. It must never be used to adjudicate which political positions are legitimate. A false dilemma or a loaded rhetorical question can serve any cause; detecting the device says nothing about the truth or worth of the position it advances. The defensive purpose of the wider programme — to measure and explain manipulation, not to rank speakers or suppress speech — depends on keeping that line bright. Released openly, the benchmark is as available to those who would audit a detector's blind spots as to those who would build one, which is the intended balance for a diagnostic instrument.

8. Conclusion and open materials

A contrastive-pairs diagnostic separates structural sensitivity from a lexical correlate in a way that corpus-level evaluation cannot. On sixty vocabulary-matched pairs, a lexical baseline sits near chance (0.633) and a contextual transformer does not (0.917, non-overlapping intervals); the margin is stable across five seeds (mean 0.880, 95% CI [0.852, 0.908]); and an independent blind rater recovers the author's labels with substantial agreement (kappa 0.617). The instrument is offered as a diagnostic with fully open materials, not as a validated measure of manipulation, and its limits define its roadmap: human annotation, tighter control phrasing on the weakest techniques, and testing on naturally occurring text. All materials — the three pair sets, the scoring harness, the per-member scores for the baseline and for every transformer seed, the independent rater's blind judgments, the multi-seed runner, and the full protocol — are released in the repository so that others can inspect, criticise, and extend them.

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